Chimpanzee-1.1: An RPS-Trained Model for ARC-AGI-3

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Abstract

This paper documents Chimpanzee-1.1, my custom-trained model for ARC-AGI-3. Chimpanzee-1.1 is a Qwen3.6-27B LoRA model trained with multimodal Direct Preference Optimization (DPO)[2] and a four-stage Regressive Plasticity Schedule (RPS). Its chosen training trajectories come from Kimi K3 solving ARC Witness and ARC Interactive games normally, without being given the correct actions. Each preference pair represents one exact pre-action state, its native image, a compact reasoning summary, and one replay-verified action. I describe the curriculum with a child-adult analogy: Level 1 examples form a high-plasticity child phase, while later-level examples form a low-plasticity adult phase. Although Chimpanzee-1.1 shares a base model and parts of its training infrastructure with Gorilla-1.1, it is meaningfully different in data-generation method, training unit, curriculum, preference construction, and context handling.

Update notice (August 15, 2026): This revision adds a recap of the ARC-AGI-1 and ARC-AGI-2 results from the original RPS study to explain the empirical motivation for applying RPS to Chimpanzee-1.1. The Chimpanzee-1.1 method, artifacts, and evaluation description are unchanged.

1. Introduction

ARC-AGI-3 is an interactive benchmark in which an agent must act inside an unfamiliar environment, infer its mechanics and objective from observations, and complete a sequence of levels.[1] Unlike static ARC tasks, the agent's actions affect what it can observe next. Training data for this setting must therefore represent not only correct outputs, but also evolving state, action effects, world-model updates, and decisions made under incomplete information.

Chimpanzee-1.1 is a model I custom-trained for this setting. More precisely, it is a multimodal LoRA adapter for Qwen3.6-27B.[4] The adapter changes the behavior of the base model through four cumulative DPO stages; it is not merely a prompting method or a renamed evaluation harness. During evaluation, I load the cumulative adapter into the Qwen3.6-27B model used by the Tufa Labs Duck harness.[5]

2. Regressive Plasticity Schedule

Regressive Plasticity Schedule (RPS) is a post-training method I created. It couples a curriculum transition with lower training plasticity: easier or foundational data is learned with a higher learning rate, while harder data is learned with a lower learning rate.[7]

2.1 Evidence From the Original RPS Study

Before Chimpanzee-1.1, I evaluated RPS on ARC-AGI-1 and ARC-AGI-2 using Qwen3-8B, LoRA, and managed DPO.[2][7] Stage 1 contained 400 easier ARC-style preference pairs and used a base learning rate of `1e-5`. Stage 2 contained 600 harder ARC-AGI-2 preference pairs and used `1e-6`. The EPS control used the same Stage 1 checkpoint, Stage 2 data, and within-stage cosine schedule, but retained a Stage 2 learning rate of `1e-5`.

On the ARC-AGI-1 public evaluation, RPS produced 17 correct test outputs out of 419, compared with 10/419 for EPS. At the task level, RPS solved 14/400 tasks and EPS solved 10/400. RPS also produced executable programs without error on 1,145/1,200 attempts, compared with 870/1,200 for EPS.

Neither schedule solved an ARC-AGI-2 public-evaluation test output; both scored 0/167. However, RPS produced executable programs without error on 234/240 attempts, compared with 188/240 for EPS. The ARC-AGI-2 result was therefore evidence about program-synthesis reliability rather than solved-task accuracy. Together, the ARC-AGI-1 accuracy and reliability gains and the ARC-AGI-2 reliability gain provided preliminary evidence of RPS's potential and motivated the Chimpanzee-1.1 experiment.

Chimpanzee-1.1 uses four named curriculum stages grouped into two plasticity stages:

Stage	Source	Difficulty rule	Preference pairs	Learning rate
Stage 1	ARC Witness	Level 1	75	`3.333333333333e-6`
Stage 1C	ARC Interactive	Level 1	75	`3.333333333333e-6`
Stage 2	ARC Witness	Later levels	144	`5e-7`
Stage 2C	ARC Interactive	Levels 2 and later	286	`5e-7`

I use a **child-adult analogy** for these two plasticity stages. Stage 1 and Stage 1C form the child stage: the model learns foundational mechanics from the first level of ARC Witness and ARC Interactive games with the higher learning rate. Stage 2 and Stage 2C form the adult stage: the model encounters later levels, where mechanics must be applied, refined, or extended, with a lower learning rate. The adult learning rate is 15% of the child learning rate. Unlike Gorilla-1.1's child-teen-adult schedule, Chimpanzee-1.1 has no separate teen stage.

The complete order is:

```
Stage 1 -> Stage 1C -> Stage 2 -> Stage 2C
```

The Stage 1 adapter is trained first. Stage 1C continues from that adapter, Stage 2 continues from the cumulative Stage 1C adapter, and Stage 2C continues from the cumulative Stage 2 adapter. The released final adapter therefore represents the complete four-stage sequence.

3. Training Dataset

3.1 Dataset Composition

The complete compact Chimpanzee-1.1 dataset contains 580 multimodal DPO pairs and 580 native pre-action PNG frames:

- 219 ARC Witness pairs from the original Chimpanzee dataset;[8]
- 361 ARC Interactive pairs added for Chimpanzee-1.1.[9]

The ARC Witness examples are divided into 75 Level 1 pairs and 144 later-level pairs. The ARC Interactive examples are divided into 75 Level 1 pairs and 286 Level 2-or-later pairs.

3.2 Normal-Solving Chosen Data

Kimi K3 generated the chosen trajectories while solving normally. It received the current multimodal state and accumulated game context, selected an action, observed the resulting state, and continued. It was not given the correct next action during this process.

For each retained action, the chosen branch contains:

1. a system prompt describing the interactive puzzle task and Python action tool;
2. the exact current 512x512 frame;

3. prior game and level context;
4. Kimi's compact reasoning summary for the next action; and
5. one replay-verified Python action call.

Raw hidden Kimi chain of thought is excluded.[3] The supervised reasoning target is the compact summary returned for that action. The action is retained only when it can be associated with the exact source state and replay-verified frame.

This per-action representation is intentional. A row does not ask the model to emit an entire level trajectory in one response. It asks for exactly one next action from an exact state. The next row can then provide the resulting state and updated context.

3.3 Rejected Data

The rejected branches were generated with Qwen3-VL-8B-Instruct. The rejection model was not given the correct action or correct reasoning. Chosen and rejected branches use the same training prompt and image.

The ARC Witness rejected branch stores the model's visible response as rejected reasoning and uses a minimal visible final output. ARC Interactive rows retain a parsed reasoning summary and proposed action when the response is valid; invalid responses use a fallback representation. The rejected branch is therefore a negative comparison target, not another replay-verified trajectory.

Many continued-stage pairs compare reasoning quality rather than action identity. In Stage 1C, 43 of 75 rejected branches choose the same action as the chosen branch. In Stage 2C, 259 of 286 do so. These pairs can still prefer a better world model or justification, but they provide less direct action-selection contrast than pairs with different actions.

3.4 Context Compaction

Long interaction histories exceeded the 11,000-token training limit. I compacted only the overlength prompt context; I did not change the image, tool schema, chosen completion, or rejected completion.

Of the 580 rows, 247 use compacted context:

Stage	Compacted rows	Recent history preserved verbatim
Stage 1	18	6 entries
Stage 1C	36	2 entries
Stage 2	55	6 entries
Stage 2C	138	2 entries

For the original ARC Witness stages, Kimi K3 compacted 72 contexts and one narrowly overlength context was manually compacted with Codex assistance. For the continued ARC Interactive stages, DeepSeek V4 Flash compacted 174 contexts with thinking enabled at high reasoning effort. The compactors did not receive the chosen action, chosen reasoning, rejected response, or current image.

Each compacted prompt separates older information into structured fields such as verified rules, verified action effects, completed-level lessons, older current-level evidence, active hypotheses, uncertainties, current objective, and failed approaches to avoid. The configured most recent history entries remain exact rather than summarized.

3.5 Dataset Validation

The final dataset passed an exact Qwen3.6/ms-swift audit before training. The audit verified all 580 rows under the training template and processor. In particular:

- every referenced PNG is present and matches its recorded hash;

- every chosen action matches its recorded source action;
- chosen and rejected prompts are identical within each pair;
- every row contains one valid multimodal image reference;
- chosen and rejected targets are nonempty;
- no branch exceeds 11,000 tokens;
- no tokens are silently truncated; and
- splitting never occurs inside a reasoning and tool-call round.

4. Training Configuration

I trained Chimpanzee-1.1 on an unquantized Qwen3.6-27B base model with native visual input. The complete configuration was:

```
Method: multimodal DPO with LoRA
Epochs: 1 per stage
Scheduler: constant_with_warmup
Warmup ratio: 0.05
Stage 1 / Stage 1C learning rate: 3.333333333333e-6
Stage 2 / Stage 2C learning rate: 5e-7
Maximum branch length: 11,000 tokens
LoRA rank / alpha / dropout: 64 / 128 / 0.05
DPO beta: 0.1
RPO alpha: 1.0
Precision: BF16
Attention implementation: SDPA
Per-device batch size: 1
Gradient accumulation: 1
Effective global batch size: 2
Distributed training: DeepSpeed ZeRO-3 across two H20 GPUs
Vision transformer: frozen
Multimodal aligner: frozen
Image-to-text conversion: none
```

Only the language-model LoRA parameters were trained. The vision transformer and multimodal aligner remained frozen. Every completed stage was exported and checksum-verified before the next stage began.

5. Evaluation Configuration

The standard Chimpanzee-1.1 evaluation uses the Tufa Duck harness and its normal interaction structure. The cumulative adapter is loaded into Qwen3.6-27B, while the harness supplies native observations, persistent interaction history, Python inspection facilities, and environment actions.

The standard evaluation does not add a proposal tournament, symbolic search system, custom memory overlay, or alternative timeout mechanism. This keeps the primary intervention in the trained model rather than introducing a second experimental harness change.

Chimpanzee-1.1 should therefore be understood as a custom-trained ARC-AGI-3 model evaluated through an existing harness. The harness remains important, but the model weights differ from vanilla Qwen3.6-27B through the cumulative LoRA adapter.

6. Difference From Gorilla-1.1

Chimpanzee-1.1 and Gorilla-1.1 share Qwen3.6-27B, multimodal LoRA DPO, native frames, and much of the same training infrastructure, but they are separate experiments. Gorilla-1.1 is documented in my earlier RPS ARC-AGI-3 report.[10]

Dimension	Chimpanzee-1.1	Gorilla-1.1
Chosen-data generation	Kimi K3 solves normally without supplied correct actions	Kimi K3 receives correct trajectories and fills in reasoning
Training unit	One exact pre-action state and one action per pair	Source-level trajectories containing multiple reasoning/action rounds
Curriculum	Four named stages grouped into child and adult plasticity stages	Three stages corresponding to child, teen, and adult plasticity stages
Context handling	Per-action context; 247 overlength prompts compacted	Uncompacted source-level trajectories, split only during training when necessary

The main conceptual difference is **normal solving versus CoT filling**. Chimpanzee-1.1 teaches decisions made while the teacher is attempting to solve the environment. Gorilla-1.1 teaches reasoning written around a supplied correct trajectory. Chimpanzee-1.1 also applies preference learning to individual action states rather than complete source-level trajectories.

A separate difference concerns Gorilla-1.1's Stage 2C prompt. During that dataset's construction, all 166 DPO pairs used one shared prompt that described the interaction as an `offline rejected-candidate call`. The same prompt was attached to the chosen branch, incorrectly describing the chosen answer as a rejected candidate. Chimpanzee-1.1 uses ordinary next-action prompts and does not contain that instruction.

These differences make Chimpanzee-1.1 a distinct ARC-AGI-3 post-training experiment rather than a continuation or repair of Gorilla-1.1.

7. Limitations

First, normal-solving data is constrained by the teacher's ability to make progress. It may contain richer decision context than CoT filling, but it cannot teach later states the teacher never reaches.

Second, 247 prompts use LLM-generated context compaction. The sampled compacted examples remain coherent and all structural checks pass, but semantic summarization cannot be guaranteed to preserve every detail. The original-stage context note also calls compaction "lossless," which is stronger than the evidence supports.

Third, three compacted Stage 2C rows leave the structured `current_objective` field empty. Their active hypotheses, recent exact history, and chosen reasoning still state the next objective, but the schema is imperfect.

Fourth, the system prompt asks the model to reason thoroughly while the per-action instruction asks it to reason normally. This is a soft stylistic tension rather than a hard action-format conflict, but it may influence response length.

Fifth, many continued-stage rejected branches choose the same action as their chosen counterparts. Those pairs primarily teach a reasoning preference and may provide a weaker action-level DPO signal.

Sixth, I trained one model family with one curriculum and one set of optimization parameters. I have not established that the same schedule transfers to other models, data sources, or interactive benchmarks.

Finally, Chimpanzee-1.1 depends on the Tufa Duck harness during evaluation. The adapter is custom-trained for ARC-AGI-3, but it is not a complete standalone agent architecture independent of that harness.

8. Solution Links

- Evaluation notebook
- Adapter on Kaggle
- Adapter on Hugging Face

- Training dataset

9. Contributions and Acknowledgments

I created the RPS idea myself and designed the Chimpanzee-1.1 experiment. I selected the normal-solving data route, curriculum stages, learning-rate schedule, and evaluation configuration. OpenAI Codex assisted me with dataset construction, compaction validation, training infrastructure, and technical auditing.[11]

The Tufa Labs authors created the Duck harness. ARC Witness and ARC Interactive contributors created the source environments. Qwen, Moonshot AI, DeepSeek, and the authors of the cited optimization methods created the underlying models and methods.

10. Conclusion

Chimpanzee-1.1 is a Qwen3.6-27B model custom-trained for ARC-AGI-3 through a four-stage multimodal DPO curriculum. Its defining choice is to learn from Kimi K3 normal-solving decisions represented as exact per-action preference pairs. RPS divides the curriculum into high-plasticity Level 1 stages and low-plasticity later-level stages. Context compaction makes long histories trainable while preserving recent transitions exactly.

Chimpanzee-1.1 shares infrastructure with Gorilla-1.1 but tests a substantially different hypothesis: whether normal-solving supervision at per-action granularity can teach an ARC-AGI-3 model more directly than reasoning filled around supplied trajectories. This report documents that approach and its limitations.

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